

Ph.D. Algebra
September 10, 1990

Time allowed: 4 hours

Read through the questions carefully before you begin. Your six best solutions will count. Theorems from the course may be quoted (within reason) as long as you give a clear indication and state the result in full.

1. Let P be a nontrivial, finite p -group, and Q a subgroup. Consider the action of P on the set P/Q of left cosets of Q by left multiplication:

$$\begin{aligned} P \times P/Q &\longrightarrow P/Q \\ (P, xQ) &\longrightarrow (px)Q. \end{aligned}$$

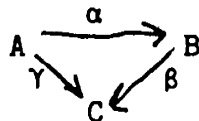
Q acts on P/Q by restriction a) Show that xQ is a fixed point of Q if and only if $x \in N_P(Q) = \{p \in P \mid pQp^{-1} = Q\}$.

b) Deduce that if Q is a proper subgroup of P then Q is also a proper subgroup of $N_P(Q)$.

c) Hence show that every maximal subgroup of P is normal of index p .

2. Let A be a Dedekind domain with field of fractions K . Show that every ideal of A can be generated by two elements.
(Hint: use the Chinese remainder theorem.)
3. Prove that any simple group of order 60 is isomorphic to A_5 .
(you need not show that A_5 is simple).

4. Let A , B and C be rings (with 1) and suppose we have a commutative diagram of ring homomorphisms:



a) Define the structure of an (A, B) bimodule on B , ie make B into a left A -module and a right B -module such that $(a.x)G = a(x.b)$ for $a \in A$, $x, b \in B$.

b) Show that for any right A -module M , we have a natural isomorphism

$$M \otimes_A C \cong (M \otimes_A B) \otimes_B C.$$

(Remember what 'natural' means in terms of the functors.

$$-\otimes_A C \text{ and } (-\otimes_A B) \otimes_B C.)$$

5. Let $\{\sigma_i\}_{i=1}^n$ be a set of n distinct automorphisms of a field

k . Show that if for $a_1, \dots, a_n \in k$

$$\sum a_i \sigma_i(u) = 0 \quad \text{for all } u \in k$$

then $a_i = 0$ for all i .

(Hint show that there is no 'shortest' relation with $a_i \neq 0$).

6. Give examples of the following (with proof).
- a) An irreducible cubic polynomial with rational coefficients whose Galois group has order 3.
 - b) Two normal extensions $K \subset L$, $L \subset M$ such that extension $K \subset M$ is not normal.
7. Let R be a ring and M simple (irreducible) R -module.
- a) Prove Schur's Lemma: every nonzero R -module map from M to itself is an isomorphism.
 - b) Give an example to show that the converse of Schur's Lemma is false.
8. Let $\mathbb{Z}_p$ denote the p -adic integers and for $n \in \mathbb{N}$
- $\epsilon_n: \mathbb{Z}_p \rightarrow \mathbb{Z}/p^n\mathbb{Z}$ the map defining the structure of $\mathbb{Z}_p$ as
- $\varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$. Let $U_n = 1 + p^n\mathbb{Z}_p$.
- a) Show that U_n is a subgroup of the group U of units of $\mathbb{Z}_p$.
 - b) Show that U_n is the kernel of $\epsilon_{n/U}: U \rightarrow (\mathbb{Z}/p^n\mathbb{Z})^*$ and that $U/U_1 \cong F_p^*$ is cyclic of order $p-1$.
 - c) For $n \geq 1$ show that the map

$1 + p^n x \mapsto x \text{ modulo } p$ defines an isomorphism

$$U_n/U_{n+1} \xrightarrow{\sim} \mathbb{Z}/p\mathbb{Z}.$$

- d) Deduce that U_1/U_n is a p -group and that U/U_n has a unique subgroup of order $p-1$.
- e) Prove that U has a subgroup of order $p-1$.
 (Hint: $U: \lim_{\leftarrow} U/U_n$)
9. Let D be a division ring and M a D -module. A subring R of $\text{End}_D(M)$ is said to be dense if for any natural number n and $v_1, \dots, v_n \in M$ which are linearly independent over D , and any n elements $w_1, \dots, w_n$ of M , there exists $r \in R$ such that $w_i = r(v_i)$ $i = 1, \dots, n$.
- a) Show that if $\dim_D M$ is finite then a dense subring of $\text{End}_D M$ must be equal to $\text{End}_D M$.
- b) Show that if M is infinite-dimensional over D and R is a dense subring of $\text{End}_D M$, then for every natural number m , R has a subring S_m which maps homomorphically onto $\text{Mat}_m(D)$.
10. a) State Hilbert's Nullstellensatz, giving the main definitions involved.
- b) Let k be an algebraically closed field. Show that every algebraic subset of k^n can be decomposed uniquely as a finite union of (irreducible) varieties.
- c) Give an example to show that in b) the union may not be disjoint.